

FPGA Synthesis and Prototyping Lab:

43

Memories

The purpose of this experiment is to Synthesis and Check the behavior of the Memories mentioned in Lab 27.

CODE:

```
module memorydemo (
input [1:0] mem_seclect ,
output reg [3:0] mem_data );

reg [3:0] mem [3:0];

wire clk ;
SB_HFOSC osc_int(
    .CLKHFPU(1'b1),
    .CLKHFEN(1'b1),
    .CLKHF(clk)
);
defparam osc_int.CLKHF_DIV = "0b10"; //12MHZ

always @(posedge clk) begin
    mem[0] <= 4'b1010;
    mem[1] <= 4'b1100;
    mem[2] <= 4'b0011;
    mem[3] <= 4'b0101;

    case (mem_seclect)
        2'b00 : mem_data <= mem[0] ;
        2'b01 : mem_data <= mem[1] ;
        2'b10 : mem_data <= mem[2] ;
        2'b11 : mem_data <= mem[3] ;
    endcase
end
endmodule
```

PCF:

```
set_io mem_seclect[0] 37
set_io mem_seclect[1] 36
set_io mem_data[0] 35
set_io mem_data[1] 34
set_io mem_data[2] 32
set_io mem_data[3] 31
```

SYNTH_COMMANDS:

```
=== memorydemo ===

Number of wires:          10
Number of wire bits:      26
Number of public wires:   7
Number of public wire bits: 23
Number of memories:       0
Number of memory bits:    0
Number of processes:      0
Number of cells:          8
  SB_DFFSR                2
  SB_DFFSS                2
  SB_HFOSC                1
  SB_LUT4                 3

2.45. Executing CHECK pass (checking for obvious problems).
checking module memorydemo..
found and reported 0 problems.

2.46. Executing JSON backend.

Warnings: 1 unique messages, 1 total
End of script. Logfile hash: 32f9405aa2
CPU: user 0.32s system 0.02s, MEM: 31.02 MB total, 25.37 MB resident
Yosys 0.9 (git sha1 1979e0b)
Time spent: 47% 10x read_verilog (0 sec), 16% 1x share (0 sec), ...
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_43$ nextpnr-ice40 --
```

IMPLIMENTATIONS

```
Info: Constraining chains...
Info:      0 LCs used to legalise carry chains.
Info: Checksum: 0x7e68b64a

Info: Annotating ports with timing budgets for target frequency 12.00 MHz
Info: Checksum: 0x4b66e1a4

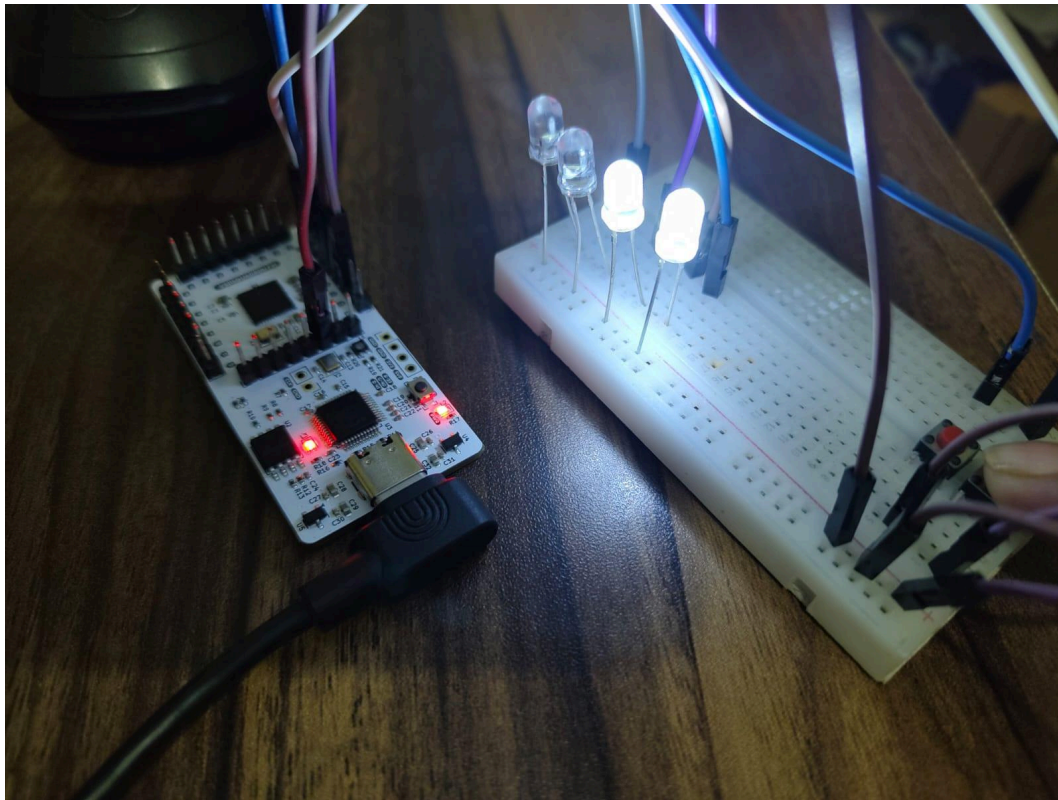
Info: Device utilisation:
Info:   ICESTORM_LC:      7/ 5280    0%
Info:   ICESTORM_RAM:     0/   30    0%
Info:   SB_IO:             6/   96    6%
Info:   SB_GB:             1/    8   12%
Info:   ICESTORM_PLL:      0/    1    0%
Info:   SB_WARMBOOT:       0/    1    0%
Info:   ICESTORM_DSP:      0/    8    0%
Info:   ICESTORM_HFOSC:    1/    1  100%
Info:   ICESTORM_LFOSC:    0/    1    0%
Info:   SB_I2C:           0/    2    0%
Info:   SB_SPI:           0/    2    0%
Info:   IO_I3C:           0/    2    0%
Info:   SB_LEDDA_IP:       0/    1    0%
Info:   SB_RGBA_DRV:       0/    1    0%
Info:   ICESTORM_SPRAM:    0/    4    0%

Info: Placed 8 cells based on constraints.
Info: Creating initial analytic placement for 6 cells, random placement wirelen
= 249.
```

PROGRAMMING

```
Info: [ 79038, 79264) |
Info: [ 79264, 79490) |
Info: [ 79490, 79716) |
Info: [ 79716, 79942) |
Info: [ 79942, 80168) |
Info: [ 80168, 80394) |*****

Info: Program finished normally.
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_43$ icepack memoryde
mo.asc memorydemo.bin
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_43$ iceprog memoryde
mo.bin
init..
cdone: high
reset..
cdone: low
flash ID: 0xEF 0x40 0x16 0x00
file size: 104090
erase 64kB sector at 0x000000..
erase 64kB sector at 0x010000..
programming..
done.
reading..
VERIFY OK
cdone: high
Bye.
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_43$
```



Video link

https://github.com/KPYerawar/NIELIT-CALICUT-FPGA-PROTOTYPING-ASSIGNMENTS/blob/main/EXP_NO_43/video.mp4

GUTHUB :

https://github.com/KPYerawar/NIELIT-CALICUT-FPGA-PROTOTYPING-ASSIGNMENTS/tree/main/EXP_NO_43